

1.1.5 匀加速运动

Motion With Constant Acceleration

$\vec{a}$ 为常矢量。设初始条件： $t = 0$ 时的位矢和速度分别为 $(\vec{r}_0, \vec{v}_0)$

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$d\vec{v} = \vec{a} dt$$

$$\int_{\vec{v}_0}^{\vec{v}} d\vec{v} = \int_0^t \vec{a} dt$$

$$\vec{v} = \vec{v}_0 + \vec{a}t \quad \left\{ \begin{array}{l} v_x = v_{0x} + a_x t \\ v_y = v_{0y} + a_y t \end{array} \right.$$

$$d\vec{r} = (\vec{v}_0 + \vec{a}t) dt$$

$$\int_{\vec{r}_0}^{\vec{r}} d\vec{r} = \int_0^t \vec{v}_0 dt + \int_0^t \vec{a}t dt$$

$$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$$

$$\left\{ \begin{array}{l} x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \\ y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2 \end{array} \right.$$

$$\vec{v} = \vec{v}_0 + \vec{a}t$$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$d\vec{r} = \vec{v} dt$$

1. 匀加速直线运动

Motion in one dimension with Constant Acceleration

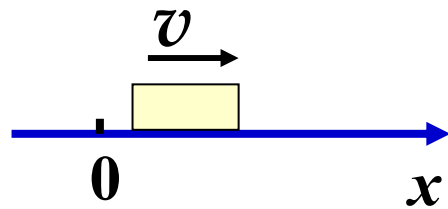
$\vec{a}$ 为常矢量, 和 $\vec{v}_0$ 在一条直线上

只用一维描述

$$\begin{cases} x = x_0 + v_0 t + \frac{1}{2} a t^2 \\ v = v_0 + a t \\ v^2 - v_0^2 = 2a(x - x_0) \end{cases}$$

若 $x_0 = 0$

$$v^2 - v_0^2 = 2ax$$

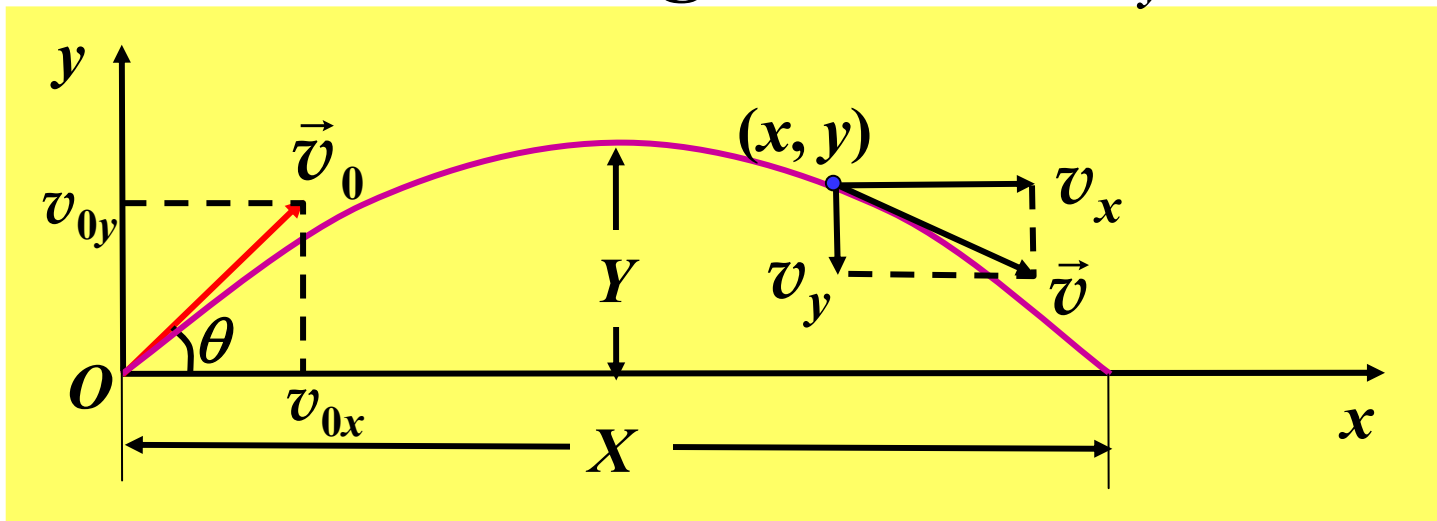


注意：式中各量可正可负



2. 抛体运动 Projectile Motion

典型的匀加速运动, $\vec{a} = \vec{g}$ $a_x = 0$, $a_y = -g$



初始条件 $x_0 = y_0 = 0$ $v_{0x} = v_0 \cos \theta$, $v_{0y} = v_0 \sin \theta$

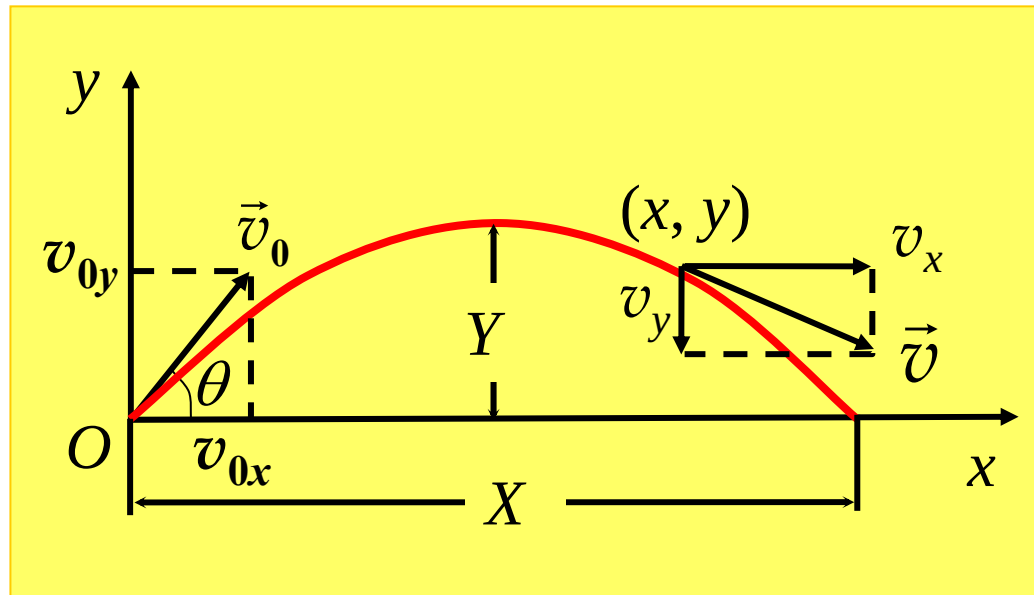
运动平面在 $(\vec{v}_0, \vec{g})$ 内.

位矢: $\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2$

$$\begin{cases} x = v_0 \cos \theta t \\ y = v_0 \sin \theta t - \frac{1}{2} g t^2 \end{cases}$$

速度: $\vec{v} = \vec{v}_0 + \vec{a} t$

$$\begin{cases} v_x = v_0 \cos \theta \\ v_y = v_0 \sin \theta - g t \end{cases}$$



轨道方程: $y = x \tan \theta - \frac{g x^2}{2 v_0^2 \cos^2 \theta}$ ——— 抛物线

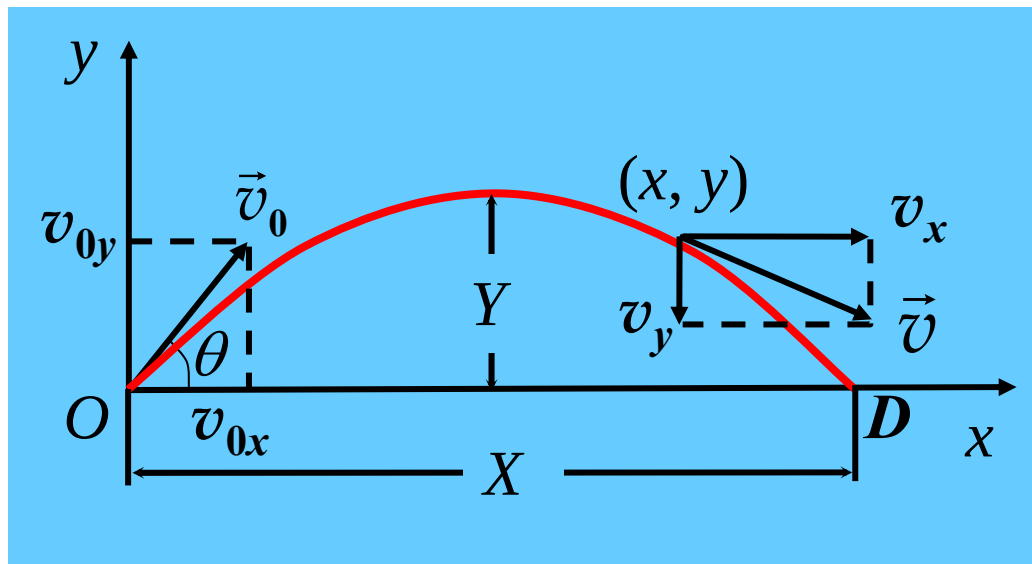
从 O 到 D 的运动时间 T

令 $y = 0$, 解得

$$T = \frac{2v_0 \sin \theta}{g}$$

射程 $X = \frac{v_0^2 \sin 2\theta}{g}$

射高 $Y = \frac{v_0^2 \sin^2 \theta}{2g}$



$$\begin{cases} x = v_0 \cos \theta t \\ y = v_0 \sin \theta t - \frac{1}{2} g t^2 \end{cases}$$